

Parameter-Efficient Fine-Tuning of Small Language Models for Turkish Legal Question Answering

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Abstract—This study investigates whether parameter-efficient supervised fine-tuning can improve small open language models for Turkish legal question answering under limited hardware resources. Two approximately three-billion-parameter instruction-tuned models, Qwen2.5-3B-Instruct and Instella-3B-Instruct, were first evaluated on the same 1,500-example test split of the Renicames/turkish-law-chatbot corpus. All baseline and final evaluations used the same Turkish legal-assistant prompt, greedy decoding, and a 128-token generation limit. Qwen was selected for adaptation because it substantially outperformed Instella on both lexical and semantic metrics. Qwen was then fine-tuned with supervised fine-tuning and QLoRA on 12,018 training examples, while 1,336 examples from the official training split were reserved for validation and the official test split remained unseen. Fine-tuning increased ROUGE-1 from 0.1886 to 0.2851, ROUGE-2 from 0.0834 to 0.1579, ROUGE-L from 0.1552 to 0.2496, and BERTScore F1 from 0.6404 to 0.7130. Qualitative analysis showed clear reductions in prompt leakage, mixed-language output, and uncontrolled repetition. However, legal hallucinations, reversed conclusions, incomplete reasoning, truncation, and dataset noise remained. The results show that QLoRA can meaningfully improve domain adaptation for Turkish legal QA, but automatic similarity metrics alone are insufficient to establish legal reliability.

Index Terms—Turkish legal NLP, question answering, small language models, supervised fine-tuning, QLoRA, LoRA, ROUGE, BERTScore

I. INTRODUCTION

Legal question answering is a demanding domain-adaptation task because a useful system must combine natural-language understanding with legal terminology, procedural knowledge, numerical conditions, and precise conclusions. Errors are particularly costly in this domain: an answer may be fluent and topically relevant while citing an incorrect article, reversing a legal condition, or inventing a sanction. Turkish legal NLP is additionally constrained by the limited availability of domain-specific resources and by the cost of adapting large models.

This project studies whether a small instruction-tuned language model can be specialized for Turkish legal question answering with limited consumer hardware. Two open models in the approximately 3B-parameter range were selected: Qwen2.5-3B-Instruct and Instella-3B-Instruct. They were evaluated on a common held-out corpus, the stronger baseline was selected, and QLoRA was used to fine-tune that model. The final contribution is an end-to-end, reproducible pipeline covering preprocessing, baseline comparison, parameter-efficient training, final evaluation, and qualitative error analysis.

The main research questions are:

- 1) Which selected 3B instruction model provides the stronger zero-shot baseline for Turkish legal QA?
- 2) How much does QLoRA-based supervised fine-tuning improve the selected model on the common test set?
- 3) Which error types remain after fine-tuning, despite improved automatic scores?

A. Related Work

Transformer language models provide the architectural basis for modern autoregressive text generation [1]. Qwen2.5 is a multilingual model family designed for instruction following and structured generation; the 3B instruction model contains approximately 3.09B parameters [4], [5]. Instella is a fully open 3B model family trained by AMD and released with weights, configurations, data information, and code [6].

Full fine-tuning becomes expensive as model size grows. LoRA freezes the original model and learns low-rank update matrices, substantially reducing the number of trainable parameters [2]. QLoRA extends this approach by propagating gradients through a frozen 4-bit quantized model into LoRA adapters. Its key memory-saving components include NF4 quantization, double quantization, and paged optimizers [3]. These properties make QLoRA suitable for this project, which was conducted on a 4 GB consumer GPU.

For evaluation, ROUGE measures lexical overlap through unigram, bigram, and longest-common-subsequence statistics [8]. Because valid legal answers may paraphrase the reference, BERTScore was also used; it aligns candidate and reference tokens through contextual embeddings rather than exact lexical matching [9]. Neither metric directly verifies legal correctness, so qualitative error analysis remains necessary.

II. EXPERIMENT SETUP

A. Dataset and Split Isolation

The experiments use the `Renicames/turkish-law-chatbot` dataset from Hugging Face [7]. The official split contains 13,354 training examples and 1,500 test examples. Only the official training split was passed to the training pipeline. The official test split was excluded from training preprocessing, validation, checkpoint selection, and hyperparameter configuration.

Exact duplicate removal was applied to the formatted training examples before splitting. No exact duplicates were found, so all 13,354 examples were retained. A deterministic 90/10 split with seed 42 produced 12,018 training examples and 1,336 validation examples. This design prevents direct test leakage and ensures that model selection after fine-tuning is based on validation loss rather than test performance.

B. Preprocessing and Prompt Format

The preprocessing code supports alternative field names for the question, optional context, and reference answer. Each example is normalized into a Turkish legal-assistant prompt:

Sistem: Sen bir Türk hukuk asistanısın. Kullanıcının hukuki sorularını doğru ve eksiksiz bir şekilde yanıtla.
Soru: <question>
Bağlam: <optional context>
Cevap: <answer>

During inference, the answer is omitted and generated by the model. The same prompt structure was used for both baselines and for the fine-tuned model.

C. Base Models

Qwen2.5-3B-Instruct is a 3.09B-parameter causal language model with 36 layers and multilingual instruction-following capabilities [5]. Instella-3B-Instruct is a text-only autoregressive 3B model with 36 decoder layers and a 4,096-token context length [6]. Both were evaluated as released, without domain adaptation, on exactly the same 1,500-item test corpus.

D. Baseline and Final Evaluation Protocol

Generation was deterministic: `do_sample=False`, no effective temperature, and `max_new_tokens=128`. Each output file stores the model identifier, question, reference answer, and generated answer. ROUGE and

TABLE I
FINE-TUNING CONFIGURATION

Parameter	Value
Base model	Qwen2.5-3B-Instruct
Method	SFT with QLoRA
Train / validation	12,018 / 1,336
Epochs	3
Per-device batch size	1
Gradient accumulation	4
Effective batch size	4
Learning rate	2×10^{-4}
Maximum sequence length	256
Random seed	42
Quantization	4-bit NF4, double quant.
LoRA rank / alpha	16 / 32
LoRA dropout	0.05
Target modules	q, k, v, o projections
Optimizer	paged AdamW 8-bit
Scheduler / warmup	linear / 0.03
Best model criterion	minimum validation loss

BERTScore were then computed from the saved outputs, avoiding a second generation pass. BERTScore used `bert-base-multilingual-cased` for all three systems.

The complete workflow is summarized in Fig. 1. Keeping the test corpus, prompt, decoding strategy, generation limit, and metric implementation fixed makes the before/after comparison controlled.

E. QLoRA Fine-Tuning

Qwen was selected for adaptation after the baseline comparison. The implementation uses Hugging Face Transformers, TRL’s `SFTTrainer`, PEFT, and bitsandbytes [10]. The base model was loaded with 4-bit NF4 quantization and double quantization. LoRA adapters were inserted into the attention projection modules `q_proj`, `k_proj`, `v_proj`, and `o_proj`. Gradient checkpointing and the paged 8-bit AdamW optimizer were used to fit training into limited GPU memory.

The successful training environment used Python 3.12.5, PyTorch 2.11.0+cu128, Transformers 5.12.0, TRL 1.6.0, and PEFT 0.19.1. Training ran on an NVIDIA GeForce RTX 3050 Ti Laptop GPU with 4 GB memory.

III. EXPERIMENT RESULTS

A. Baseline Model Selection

Table II presents all test-set results. Qwen strongly outperformed Instella on every metric. Its ROUGE-1 was approximately 7.25 times larger, while BERTScore F1 increased from 0.5249 to 0.6404. Instella’s extremely low ROUGE-2 indicates that its responses rarely reproduced reference bigrams, and qualitative inspection showed frequent malformed or irrelevant text. Qwen was therefore selected for fine-tuning.

Same prompt and deterministic decoding

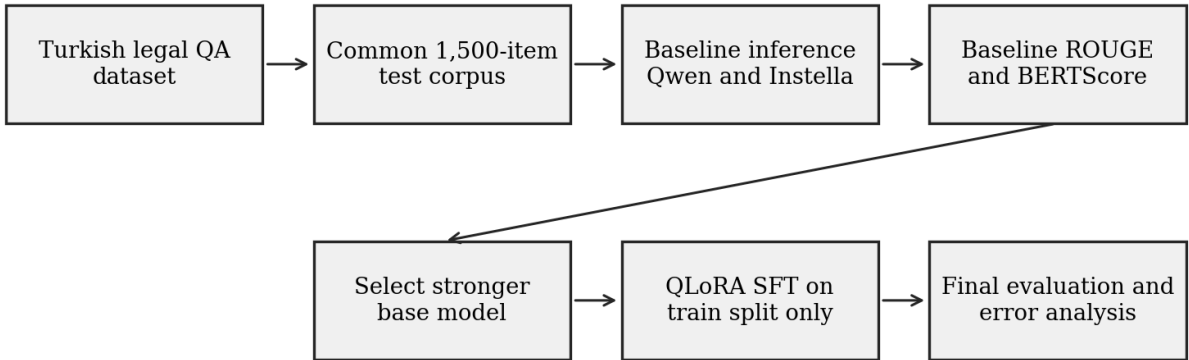


Fig. 1. Experimental workflow. The official test split is shared by all evaluations but is never used during training or validation.

TABLE II
RESULTS ON THE COMMON 1,500-EXAMPLE TEST CORPUS

System	R-1	R-2	R-L	R-Lsum	BERT-P	BERT-R	BERT-F1
Instella-3B-Instruct baseline	0.0260	0.0004	0.0229	0.0233	0.5136	0.5403	0.5249
Qwen2.5-3B-Instruct baseline	0.1886	0.0834	0.1552	0.1544	0.6230	0.6612	0.6404
Qwen2.5-3B-Instruct + QLoRA	0.2851	0.1579	0.2496	0.2500	0.7095	0.7189	0.7130

B. Before/After Fine-Tuning

Fine-tuning improved every reported metric. Relative to the Qwen baseline, ROUGE-1 increased by 51.18%, ROUGE-2 by 89.39%, ROUGE-L by 60.78%, and ROUGE-Lsum by 61.96%. BERTScore precision increased by 13.89%, recall by 8.73%, and F1 by 11.35%. The larger relative gains in ROUGE than in BERTScore suggest that fine-tuning primarily made the responses more aligned with the dataset’s terminology, phrasing, and answer structure, while also producing a meaningful semantic improvement.

Figure 2 highlights the progression from Instella to Qwen and from baseline Qwen to fine-tuned Qwen. The improvement is not limited to one metric, which reduces the likelihood that the conclusion is an artifact of a single evaluation measure.

C. Training and Validation Behavior

The final reported training loss was 0.6016 and the final validation loss was 0.5363. Validation mean token accuracy reached 0.8768. The total training runtime was 44,380.9 seconds, or approximately 12.33 hours. The validation loss being lower than the aggregate training loss does not by itself indicate leakage: training loss is accumulated while dropout and optimization noise are active, whereas vali-

dation is measured in evaluation mode. More importantly, the official test split was not used during this process.

The strong final test improvement is consistent with successful domain adaptation. Nevertheless, no multi-seed experiment or hyperparameter sweep was conducted, so the results describe one controlled training run rather than a statistically estimated distribution.

D. Error Analysis

Automatic scores improved, but qualitative comparison reveals important remaining risks. The analysis was performed on matching baseline and fine-tuned outputs from the common test corpus. Table III summarizes representative cases.

The dominant observed categories are:

- **Prompt leakage and role confusion:** common in the baseline and substantially reduced after fine-tuning.
- **Degenerate repetition:** severe in several baseline answers; reduced but not eliminated.
- **Legal hallucination:** invented provisions, offense labels, penalties, or procedures remain in fluent fine-tuned outputs.
- **Reversed conclusions:** the model may identify the relevant article but return the opposite yes/no decision.

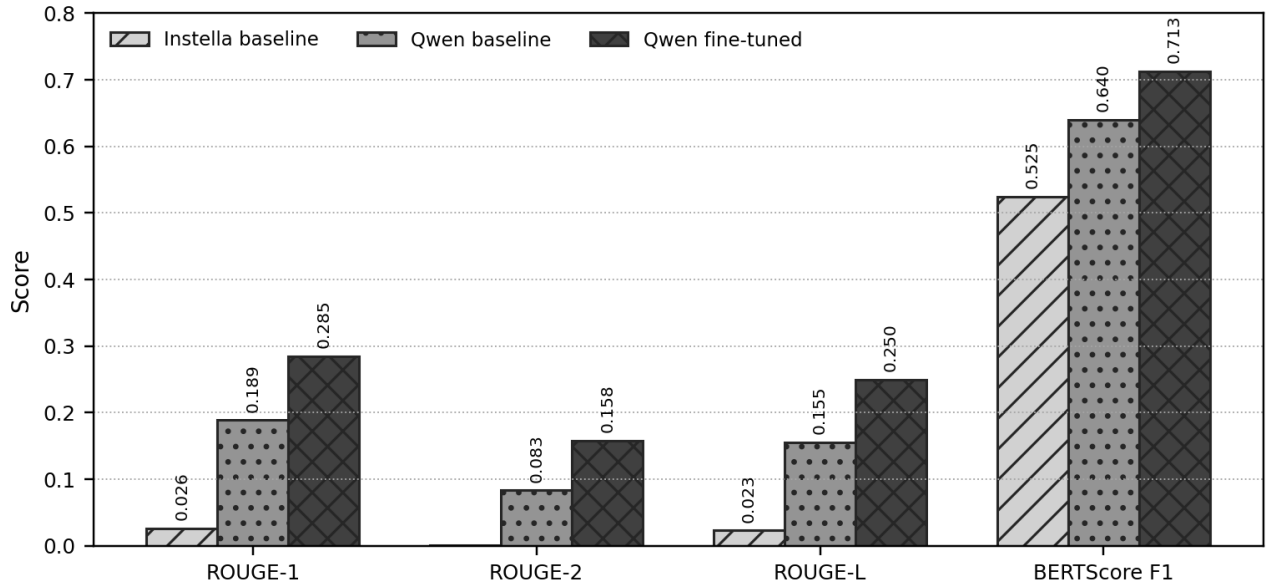


Fig. 2. Comparison of selected lexical and semantic metrics. All values are computed on the same 1,500-example test set.

TABLE III
REPRESENTATIVE QUALITATIVE ERRORS

Question type	Baseline behavior	Fine-tuned behavior and interpretation
Constitution Art. 90	Incorrect claim, English fragment, and reproduction of the system/question prompt.	Prompt leakage disappears, but the response mostly restates the question and omits the treaty-approval condition. Formatting improves more than legal completeness.
Constitution Art. 94	Off-topic continuation and another generated prompt.	Directly addresses the article but states that the application does not conflict, reversing the reference conclusion.
Constitution Art. 102	Invents contradictory numerical rules for the election period.	Remains incorrect by treating the 70-day condition as compatible with the provision. Article recognition is not equivalent to legal reasoning.
Financial-advice fraud	Produces a meaningless token followed by prompt repetition.	Becomes fluent, but invents offense names and an unsupported 10–30 year penalty. This is a high-risk legal hallucination.
Public officials / Art. 129	Degenerate repetition until the generation limit.	More coherent and less repetitive, yet still omits the reference conditions such as loyalty to law, impartiality, and public service.
“Beyanname nedir?”	The dataset question and reference answer are semantically inconsistent.	The answer is also poor, but automatic scoring is unreliable because the reference defines VAT rather than a declaration.

- **Incomplete answers and truncation:** some responses omit decisive conditions or end mid-sentence at the 128-token limit.
- **Dataset noise:** mismatched questions and references can penalize otherwise plausible answers and distort automatic metrics.

These observations explain why higher ROUGE and BERTScore values should be interpreted as improved similarity to the corpus, not as proof of legal correctness. In legal QA, expert review or fact-grounded evaluation is necessary before deployment.

E. Threats to Validity and Limitations

The experiment has five main limitations. First, only one fine-tuning seed and one main hyperparameter configuration were evaluated. Second, ROUGE and BERTScore cannot verify whether a statute, time limit, or sanction is legally correct. Third, the qualitative analysis is representative rather than an exhaustive expert annotation of all 1,500 answers. Fourth, the 128-token generation limit may truncate detailed responses. Fifth, the dataset contains at least some noisy or mismatched references. The trained model must therefore be treated as an academic prototype and not as a source of professional legal advice.

IV. CONCLUSION

This project compared two small instruction-tuned models and applied parameter-efficient domain adaptation to Turkish legal question answering. On the common 1,500-example test corpus, Qwen2.5-3B-Instruct clearly outperformed Instella-3B-Instruct and was selected for QLoRA fine-tuning. Training used 12,018 examples, a separate 1,336-example validation set, and no access to the official test set.

Fine-tuning improved all lexical and semantic metrics: ROUGE-1 rose from 0.1886 to 0.2851, ROUGE-2 from 0.0834 to 0.1579, ROUGE-L from 0.1552 to 0.2496, and BERTScore F1 from 0.6404 to 0.7130. The qualitative results also show better instruction adherence, less prompt leakage, and reduced uncontrolled repetition. However, fluent legal hallucinations, wrong yes/no conclusions, incomplete explanations, truncation, and dataset noise remain.

The central conclusion is that QLoRA provides an effective and hardware-efficient method for adapting a 3B multilingual model to Turkish legal QA. At the same time, similarity-based evaluation must be complemented by legal expert review, retrieval grounding, citation verification, and uncertainty-aware refusal behavior before such a system could be considered reliable. Future work should include multi-seed experiments, answer-only loss masking, retrieval-augmented generation from authoritative Turkish legislation, systematic human evaluation, and automatic

verification of cited articles and numerical legal conditions.

ETHICAL AND LEGAL NOTICE

This study is intended for academic research. The generated answers may be incorrect, incomplete, or outdated and do not constitute legal advice.

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